

3 pH water chemistry and irrigation

These factors control nutrient solubility, salt exposure, water entry, and the soil volume explored by roots.

pH is a gatekeeper not a nutrient

- **What pH tells us.** It describes how acidic or alkaline the soil solution is and helps predict which nutrients may become less available.
- **High pH.** Iron, zinc, manganese, and phosphorus can become difficult for roots to obtain even when the soil contains them.
- **Low pH.** Some metals become more soluble, while calcium, magnesium, and microbial processes may be limited.
- **Roots can imitate a pH problem.** Wet, diseased, or poorly aerated roots may fail to take up nutrients, worsening chlorosis that appears chemical.
- **Test the active root zone.** A bulk soil sample can hide differences between the wetted area, dry edges, surface mulch, and deeper layers.

Three water-related problems that are often confused

Problem	What it does	Useful measurements
Salinity	Total dissolved salts make water harder for roots to extract. Plants can appear drought-stressed even when soil is moist.	Water EC soil saturation extract E _{Ce} and seasonal trend
Sodicity	Too much sodium relative to calcium and magnesium can disperse clay and reduce infiltration and aeration.	Water and soil SAR or exchangeable sodium plus infiltration behavior
Specific ion toxicity	Chloride sodium or boron can accumulate in sensitive tissue and cause direct injury.	Water soil and leaf concentration of the specific ion

Critical distinction

Leaching removes soluble salts only when water can pass below the active roots and leave the site. If drainage is restricted, simply applying more water can worsen low oxygen and root disease.

Irrigation regime means more than total gallons

- **Amount.** Replace crop water use and provide only the justified extra water needed for salt control.
- **Frequency.** Keep shallow avocado feeder roots from cycling between severe drying and prolonged saturation.
- **Duration and rate.** Wet the intended root depth at a rate the soil can take in, without runoff or unnecessary deep loss.
- **Uniformity.** Low-output areas can remain dry and salty while high-output areas remain wet.
- **Wetted pattern.** A narrow wet bulb limits root volume and concentrates salts at its edges.
- **Seasonal leaching.** Track rainfall, water quality, and drainage; dry winters may provide less natural salt removal.